

"Kestrel 2.0 Hybrid: Headlamp condensation and LED failure requires assembly"

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Area: bulletins

1 Condition

Owners of the Kestrel 2.0 Hybrid may report persistent internal condensation behind the headlamp lens accompanied by intermittent or total loss of the low-beam LED array. The affected unit is the integrated LED headlamp assembly carrying part number HLMP-7020, which is sealed and not field-serviceable at the optic or driver-board level. Inspect both left and right HLMP-7020 units, as moisture intrusion frequently affects only one side initially. See the Kestrel Body Service Manual :: Lighting Sub-System for the full lighting circuit description before beginning diagnosis.

2 Cause

The root cause is a compromised perimeter seal on the HLMP-7020 assembly that allows humid air to enter and condense against the cooler inner lens surface. Repeated thermal cycling of the LED driver then accelerates moisture migration onto the driver board, producing open or short conditions that disable the low-beam array. Because the HLMP-7020 housing is ultrasonically welded, the seal failure cannot be repaired and the complete assembly must be replaced. Refer to the Kestrel Body Service Manual :: Body Electrical System Overview for grounding and connector context that can mimic these symptoms.

3 Repair Procedure

Replace the affected HLMP-7020 headlamp assembly following PROC-HEADLAMP, which covers fender liner access, connector release, and aiming after installation. Torque the three mounting studs to specification and verify the low-beam array energizes without flicker before releasing the vehicle. Complete details are in the Kestrel Body Service Manual :: Headlamp Assembly Replacement Procedure , and the parent Kestrel Body Service Manual provides aiming targets. If diagnostic codes point upstream of the assembly, consult the Kestrel Electrical Service Manual for harness and module checks before condemning the HLMP-7020 unit.